

Transforming and Cleaning Data



Mohit Batra

DATA ENGINEER

[linkedin.com/in/mohitbatra](https://www.linkedin.com/in/mohitbatra)



Overview



Understand the dimensional model

Compare common transformations in other ETL tools with Spark

Analyze source data

Clean it by applying data quality checks

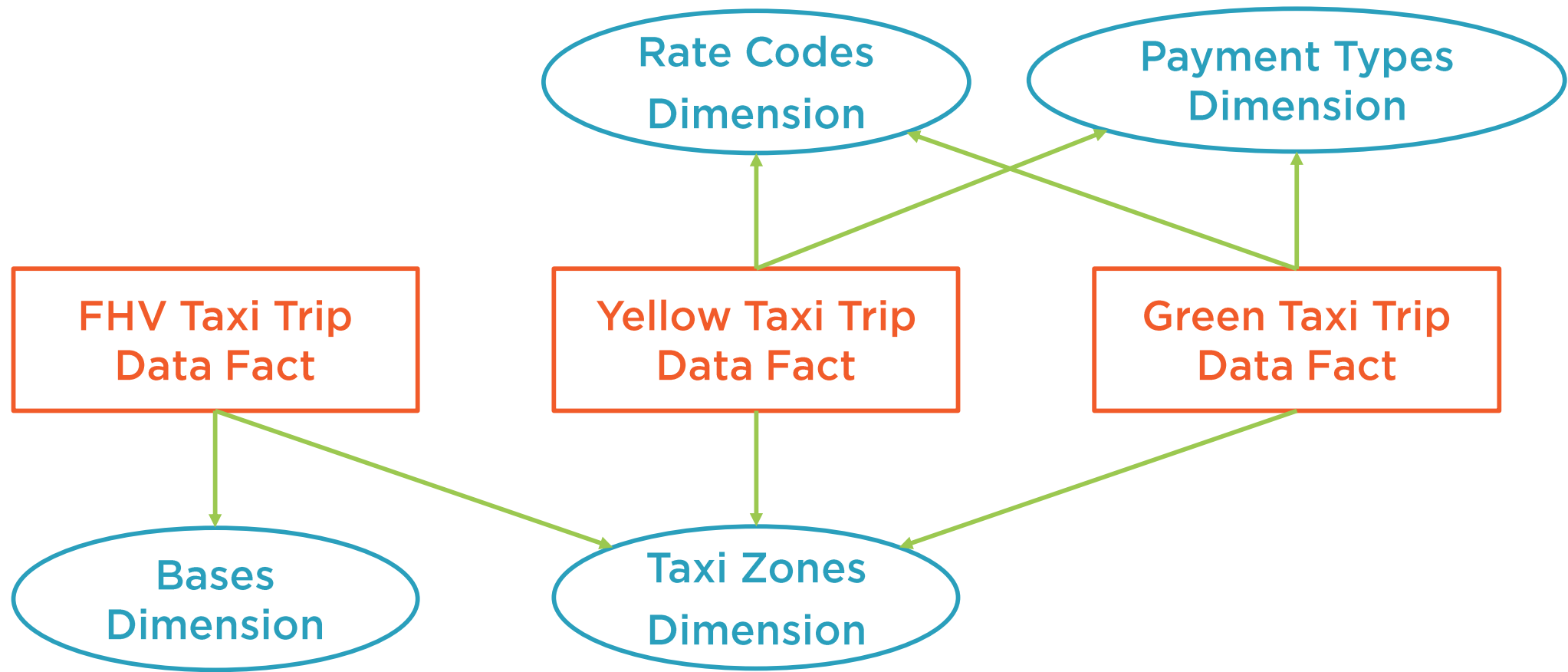
Apply business transformations in Scala to build model

Work with Spark SQL

Handle corrupt data



Dimensional Model



Understanding Common Transformations



dataframe

`.where(<expression>)`

SELECT ...

WHERE <expression>

Filter Data

SSIS | Conditional Split

Informatica | Filter



dataframe

```
.withColumn("<col name>", <expression>)
```

```
SELECT <expression> AS <col name>
```

Calculated Columns

SSIS | Derived Column

Informatica | Expression



```
dataframe
```

```
.count()
```

```
SELECT COUNT(*)
```

Count Rows

SSIS | Row Count

Informatica | Aggregator



dataframe

```
.groupBy(<columns to group on>)
```

```
.agg(<aggregation expression>)
```

```
SELECT <aggregation expression>
```

```
GROUP BY <columns to group on>
```

Aggregate Data

SSIS | Aggregate

Informatica | Aggregator



dataframe

`.orderBy(<columns to sort on>)`

SELECT ...

ORDER BY <columns to sort on>

Sort Dataset

SSIS | Sort

Informatica | Sorter



dataframe1

```
.join(dataframe2, <join condition>, <join type>)
```

SELECT ...

FROM table1

<join type> JOIN table2 <join condition>

Join 2 Datasets

SSIS | Merge Join

Informatica | Joiner



```
dataframe1
```

```
.union(dataframe2)
```

```
SELECT ... FROM table1
```

```
UNION
```

```
SELECT ... FROM table2
```

Union 2 Datasets

SSIS | Merge or Union All

Informatica | Union



Analyzing and Cleaning Data



Data Quality Checks

Completeness

- Remove null or missing data
- Fill missing values with placeholder

Uniqueness

- Remove duplicate records

Timeliness

- Appropriate date range

Accuracy

- Remove inaccurate data



Applying Transformations



Working with Spark SQL



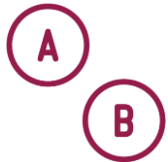
Execution Context



Execution context is an isolated environment in which code is executed, and state is maintained (variables, objects, functions)



In Databricks, each language and notebook combination has a separate execution context on a cluster



Being isolated, objects in one execution context cannot be shared with other execution context



To work with multiple languages and notebooks, pass around the data from one execution context to another



Handling Corrupt Data



Options to Handle Corrupt Data

TextFile formats (CSV and JSON) has built-in support for handling corrupt records

Support 3 ParseModes while reading data

- Permissive
- DropMalformed
- FailFast

Databricks allows for storing corrupt records & reasons at user defined location



Summary



Dimensional model for Taxi Data

Compared common transformations in SSIS, Informatica, Scala and Spark SQL

Used *describe* to analyze data

Apply data quality checks and clean up

- drop, fill, dropDuplicates, filter...

Applied transformations

- select, joins, groupBy, agg, orderBy...

Worked with Spark SQL

- Temp views – Local and Global

Handle corrupt data

- Modes, BadRecordsPath

